



Empowering Decisions with BI: Analyzing Retail Chain Performance

Tools: Excel • PostgreSQL • Tableau | Analyst: Patrick Chowthi

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Stock ID: 47857043



Introduction

Introduction

Background / Context

- This analysis examines the performance of a retail chain that sells supermarket goods across multiple cities.
- The company aims to better understand its **sales patterns**, **customer preferences**, and **product performance** to drive profitability.

Project Goal

Use data analysis and visualization tools — *Excel, PostgreSQL, and Tableau* — to uncover actionable insights that will help management:

- Identify **top-performing products and regions**
- Recognize **customer behavior trends**
- Optimize **sales and inventory strategies**



Business Scenario and Objectives

Content:

The retail chain operates multiple branches in various cities, offering a wide range of consumer goods. Facing growing competition, the organization seeks to leverage its transactional sales data to make data-driven decisions

Objective of the Analysis:

- Assess overall sales performance across regions
- Evaluate product category trends and profitability
- Identify key drivers of Sales through statistical and visual analysis



Methodology

Data Details

The project started with multiple raw datasets in CSV excel format.

The CSV files were sales, store_names, store_cities, product_hierarchy, product_names and city_names

- sales- transactional sales data
- store_names and store_cities- Store identifiers and locations
- product_hierarchy and product_names- product categories and names
- city_names- City identifiers
- Data required cleaning and preparation to ensure consistency, completeness, and usability for analysis.

Module 1: Data Cleaning and Preparation Using Excel

Objective- Clean and prepare the downloaded raw data in order to ensure accuracy and consistency for deep analysis

Brief Overview:

- 1) Imported CSV files into Excel and examined dataset structure
- 2) Cleaned data by checking for missing values and duplicate data
- 3) Standardize date and text format across tables using trim and date functions
- 4) Prepared raw data set into clean data set for deeper analysis using pivot tables

Tools Used: Microsoft Excel, Pivot Tables

Module 2: Data Querying and Analysis Using PostgreSQL

Objective: Perform multidimensional analysis using SQL to uncover trends across stores, regions, and products

Key Steps:

- Loaded cleaned datasets into Postgre SQL database
- Created SQL queries for table verification and data aggregation
- Created ROLLUP and CUBE queries for hierarchical summaries
- Explored sales trends by time, region and product hierarchy

Tools used: Postgre SQL, pgadmin

Module 3: Data Visualization and Statistical Analysis

Objective: Visualize key sales trends and evaluate relationships in the data.

Key Steps:

- Created line, bar, sunburst, and scatter charts in Excel
- Performed linear regression to forecast trends and assess correlations
- Explored relationships between sales, stock, and dates

Tools Used: Microsoft Excel, Charts, Regression Analysis

Module 4: Data Visualization and Dashboards Using Tableau

Objective: Build an interactive dashboard integrating previous analyses for decision-making insights.

Key Steps:

- Created multiple worksheets with charts and KPIs
- Combined worksheets into an interactive Tableau dashboard
- Implemented filters, calculated fields, and interactive visual elements

Tools Used: Tableau Desktop



Results

Module 1, Lesson 1: Data Cleaning and Preparation

Before

The screenshot displays the Microsoft Excel application window. The title bar shows the file name 'BI_Analyst_Capstone_M01L01'. The ribbon includes tabs for File, Home, Insert, Share, Page Layout, Formulas, Data, Review, View, Help, and Draw. The search bar at the top right contains the text 'Search for tools, help, and more (Alt + Q)'. The status bar at the bottom left shows 'Workbook Statistics'.

The data table is as follows:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
182	P0017	S0027	2/1/2017	1	1.38	49	1.49	PR14		PR03					
183	P0017	S0028	2/1/2017	5	4.63	0		PR14		PR03					
184	P0017	S0031	2/1/2017	0	0	40	1.49	PR14		PR03					
185	P0017	S0035	2/1/2017	1	1.38	21	1.49	PR14		PR03					
186	P0017	S0043	2/1/2017	1	1.38	29	1.49	PR14		PR03					
187	P0017	S0045	2/1/2017	0	0	25	1.49	PR14		PR03					
188	P0017	S0048	2/1/2017	9	8.33	0		PR14		PR03					
189	P0017	S0049	2/1/2017	0	0	44	1.49	PR14		PR03					
190	P0017	S0050	2/1/2017	0	0	41	1.49	PR14		PR03					
191	P0017	S0051	2/1/2017	1	1.38	17	1.49	PR14		PR03					
192	P0017	S0052	2/1/2017	1	1.38	13	1.49	PR14		PR03					
193	P0017	S0055	2/1/2017	0	0	46	1.49	PR14		PR03					
194	P0017	S0056	2/1/2017	1	1.38	23	1.49	PR14		PR03					
195	P0017	S0058	2/1/2017	2	2.76	49	1.49	PR14		PR03					
196	P0017	S0060	2/1/2017	0	0	24	1.49	PR14		PR03					
197	P0017	S0062	2/1/2017	2	2.76	47	1.49	PR14		PR03					
198	P0017	S0063	2/1/2017	0	0	2	1.49	PR14		PR03					
199	P0017	S0066	2/1/2017	1	1.38	16	1.49	PR14		PR03					
200	P0017	S0067	2/1/2017	0	0	9	1.49	PR14		PR03					
201	P0017	S0070	2/1/2017	0	0	49	1.49	PR14		PR03					
202	P0017	S0072	2/1/2017	5	4.63	0		PR14		PR03					

The 'Conditional Formatting' pane on the right shows the 'Add a formatting rule' section with a 'New Rule' button. The status bar at the bottom shows 'Workbook Statistics' and a zoom level of 100%.

Module 1, Lesson 1: Data Cleaning and Preparation

After

=XLOOKUP(A17,Product_Hierarchy1:A,Product_Hierarchy1:B,"Not found")																										
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	
	product_id	store_id	date	sales	revenue	stock	price	promo_type_1	promo_bin_1	promo_type_2	promo_bin_2	promo_discount_2	promo_discount_type_2	Store name	Store id	Store size	Product name	Product length	Product depth	Product width	cluster_id	hierarchy1_id	hierarchy2_id	hierarchy3_id	hierarchy4_id	
1	P0005	S0001	1/2/2017	0	0	7	33.9 PR14			PR03				Electro World (National Chain)	S0001	41	Wine Cooler	16	30	16	cluster_9	H03	H0312	H031211	H03121109	
2	P0011	S0001	1/2/2017	0	0	10	49.9 PR14			PR03				Electro World (National Chain)	S0001	41	Induction Cooktop	9	22	15	cluster_0	H03	H0315	H031506	H03150612	
3	P0015	S0001	1/2/2017	1	2.41	20	2.6 PR14			PR03				Electro World (National Chain)	S0001	41	Convection Oven	10	33	10	cluster_1	H00	H0000	H000003	H00000309	
4	P0017	S0001	1/2/2017	0	0	13	1.49 PR14			PR03				Electro World (National Chain)	S0001	41	Pizza Oven	8	28	8	cluster_6	H00	H0000	H000003	H00000310	
5	P0018	S0001	1/2/2017	0	0	49	1.95 PR14			PR03				Electro World (National Chain)	S0001	41	Toaster Oven	1	14	11	cluster_4	H00	H0003	H000316	H00031609	
6	P0024	S0001	1/2/2017	0	0	23	1.95 PR14			PR03				Electro World (National Chain)	S0001	41	Stackable Washer and Dryer	1	23	16	cluster_0	H03	H0314	H031404	H03140407	
7	P0035	S0001	1/2/2017	2	4.54	14	2.45 PR14			PR03				Electro World (National Chain)	S0001	41	Immersion Blender	3	17	12.5	cluster_7	H00	H0003	H000311	H00031100	
8	P0046	S0001	1/2/2017	0	0	4	34.5 PR14			PR03				Electro World (National Chain)	S0001	41	Smart Doorbells (with video)	1.7	11.1	9.6	cluster_0	H01	H0108	H010809	H01080900	
9	P0051	S0001	1/2/2017	0	0	27	0.7 PR14			PR03				Electro World (National Chain)	S0001	41	Smart Speakers (with voice assistants)	1.7	17.5	4.5	cluster_7	H00	H0003	H000314	H00031409	
10	P0054	S0001	1/2/2017	0	0	6	3.95 PR14			PR03				Electro World (National Chain)	S0001	41	Robot Vacuums	8.5	14.6	8.5	cluster_9	H00	H0002	H000201	H00020110	
11	P0055	S0001	1/2/2017	0	0	14	3.5 PR05			PR03				Electro World (National Chain)	S0001	41	Smart TVs	2.3	18.5	13.5	cluster_0	H00	H0003	H000311	H00031109	
12	P0057	S0001	1/2/2017	0	0	6	12.9 PR14			PR03				Electro World (National Chain)	S0001	41	Upright Vacuums	4	22	9	cluster_0	H01	H0108	H010807	H01080709	
13	P0060	S0001	1/2/2017	0	0	1	15.9 PR14			PR03				Electro World (National Chain)	S0001	41	Steam Mops	7.5	20.5	5.4	cluster_0	H01	H0106	H010600	H01060013	
14	P0062	S0001	1/2/2017	0	0	1	19.9 PR14			PR03				Electro World (National Chain)	S0001	41	Hardwood Floor Cleaners	0	0	0	cluster_0	H03	H0312	H031205	H03120507	
15	P0066	S0001	1/2/2017	0	0	43	2.5 PR14			PR03				Electro World (National Chain)	S0001	41	Electric Grills	0.3	15.5	8.5	cluster_0	H03	H0313	H031305	H03130515	
16	P0067	S0001	1/2/2017	0	0	4	15.9 PR14			PR03				Electro World (National Chain)	S0001	41	Smokers	5.4	20.5	7.5	cluster_4	H01	H0106	H010600	H01060013	
17	P0068	S0001	1/2/2017	0	0	6	5.1 PR10			PR03				Electro World (National Chain)	S0001	41	Patio Heaters	13	9	13	cluster_8	H00	H0000	H000003	H00000300	
18	P0070	S0001	1/2/2017	0	0	1	6.25 PR14			PR03				Electro World (National Chain)	S0001	41	Lawn Mowers (gas, electric)	8.7	14	8.7	cluster_0	H00	H0002	H000201	H00020100	
19	P0071	S0001	1/2/2017	0	0	1	9.45 PR14			PR03				Electro World (National Chain)	S0001	41	String Trimmers	4.9	16.5	6.5	cluster_0	H01	H0108	H010807	H01080700	
20	P0072	S0001	1/2/2017	0	0	13	3.9 PR14			PR03				Electro World (National Chain)	S0001	41	Leaf Blowers	3	12	12	cluster_0	H03	H0314	H031402	H03140200	
21	P0079	S0001	1/2/2017	0	0	46	2.25 PR14			PR03				Electro World (National Chain)	S0001	41	Phone Chargers	2	13	6	cluster_5	H00	H0003	H000316	H00031600	
22	P0080	S0001	1/2/2017	0	0	10	7.9 PR06			PR03				Electro World (National Chain)	S0001	41	Portable Speakers	2	24	2	cluster_0	H03	H0312	H031204	H03120400	
23	P0083	S0001	1/2/2017	0	0	2	6.9 PR05			PR03				Electro World (National Chain)	S0001	41	Security Cameras	2.2	7.7	10.6	cluster_6	H00	H0002	H000201	H00020109	
24	P0089	S0001	1/2/2017	0	0	2	5.99 PR14			PR03				Electro World (National Chain)	S0001	41	Portable Blenders	2	15	8	cluster_0	H03	H0315	H031512	H03151215	
25	P0090	S0001	1/2/2017	0	0	6	2.9 PR14			PR03				Electro World (National Chain)	S0001	41	Handheld Vacuum Cleaner	1	5	6	cluster_0	H03	H0312	H031211	H03121109	
26	P0092	S0001	1/2/2017	0	0	20	1.55 PR14			PR03				Electro World (National Chain)	S0001	41	Canister Vacuum Cleaner	6	15	8	cluster_6	H00	H0001	H000102	H00010210	
27	P0099	S0001	1/2/2017	0	0	19	10.9 PR14			PR03				Electro World (National Chain)	S0001	41	Juicer	3	11.3	10	cluster_0	H03	H0313	H031302	H03130210	
28	P0103	S0001	1/2/2017	13	31.9	52	2.65 PR14			PR03				Electro World (National Chain)	S0001	41	Handheld Milk Frother	9	30	9	cluster_0	H00	H0000	H000003	H00000300	
29	P0109	S0001	1/2/2017	0	0	19	8.95 PR14			PR03				Electro World (National Chain)	S0001	41	Hair Straightener	7	18	11	cluster_0	H01	H0107	H010700	H01070009	
30	P0110	S0001	1/2/2017	1	2.73	4	2.95 PR14			PR03				Electro World (National Chain)	S0001	41	Electric Shaver	6.2	16.8	6.2	cluster_0	H00	H0000	H000004	H00000405	
31	P0114	S0001	1/2/2017	0	0	12	0.45 PR14			PR03				Electro World (National Chain)	S0001	41	LED TV	2	7.5	9	cluster_0	H00	H0003	H000312	H00031209	
32	P0116	S0001	1/2/2017	0	0	1	0.85 PR14			PR03				Electro World (National Chain)	S0001	41	Smart TV	0.5	14	6	cluster_3	H00	H0002	H000200	H00020009	
33	P0125	S0001	1/2/2017	0	0	22	2.45 PR14			PR03				Electro World (National Chain)	S0001	41	Dehumidifier	2	5	9	cluster_5	H00	H0003	H000316	H00031600	
34	P0129	S0001	1/2/2017	0	0	10	24.95 PR14			PR03				Electro World (National Chain)	S0001	41	Portable Heater	22.2	22.2	44.4	cluster_0	H01	H0107	H010701	H01070110	
35	P0131	S0001	1/2/2017	0	0	28	2.25 PR14			PR03				Electro World (National Chain)	S0001	41	Counter-depth Refrigerator	1	12	11	cluster_6	H00	H0003	H000316	H00031609	
36	P0132	S0001	1/2/2017	0	0	2	7.99 PR14			PR03				Electro World (National Chain)	S0001	41	Smart Refrigerator (Wi-Fi enabled)	5	28	16	cluster_0	H03	H0311	H031109	H03110901	
37	P0137	S0001	1/2/2017	0	0	8	23.9 PR14			PR03				Electro World (National Chain)	S0001	41	Chest Freezer with Lock and Key	23.6	24.1	8.3	cluster_0	H03	H0315	H031502	H03150200	
38	P0140	S0001	1/2/2017	2	11	13	5.5 PR14			PR03				Electro World (National Chain)	S0001	41	Self-Cleaning Oven	2	27	21	cluster_0	H03	H0313	H031307	H03130700	
39	P0144	S0001	1/2/2017	0	0	4	4.95 PR10			PR03				Electro World (National Chain)	S0001	41	Countertop Pizza Oven with Rotisserie F	4.5	21	14.3	cluster_3	H00	H0003	H000311	H00031100	
40	P0147	S0001	1/2/2017	0	0	6	3.99 PR14			PR03				Electro World (National Chain)	S0001	41	High-Capacity Washing Machine	12.5	37	12.5	cluster_9	H01	H0106	H010601	H01060112	
41	P0148	S0001	1/2/2017	4	21.3	5	5.75 PR14			PR03				Electro World (National Chain)	S0001	41	Steam Washer with Sanitize Cycle	6	7	12.2	cluster_0	H00	H0002	H000200	H00020000	
42	P0150	S0001	1/2/2017	0	0	6	12.99 PR14			PR03				Electro World (National Chain)	S0001	41	Stackable Washer and Dryer with Steam	6	14	14	cluster_0	H03	H0313	H031302	H03130209	
43	P0151	S0001	1/2/2017	0	0	2	5.9 PR14			PR03				Electro World (National Chain)	S0001	41	Multicooker	2	15	11.5	cluster_0	H01	H0108	H010806	H01080600	
44	P0157	S0001	1/2/2017	0	0	5	6.95 PR14			PR03				Electro World (National Chain)	S0001	41	Immersion Blender with Soup Maker Att	6	6	15	cluster_2	H00	H0003	H000312	H00031200	
45	P0160	S0001	1/2/2017	0	0	12	2.5 PR14			PR03				Electro World (National Chain)	S0001	41	High-Speed Blender	30	1	21	cluster_0	H03	H0313	H031305	H03130520	
46	P0162	S0001	1/2/2017	0	0	9	3.9 PR14			PR03				Electro World (National Chain)	S0001	41	Electric Griddle with Nonstick Surface	2	8.8	3.6	cluster_0	H01	H0108	H010810	H01081009	
47	P0164	S0001	1/2/2017	0	0	7	24.9 PR14			PR03				Electro World (National Chain)	S0001	41	Robot Vacuum with Self-Empty Base	4.4	15.8	9	cluster_0	H01	H0108	H010807	H01080700	

Line Graph

Consolidated

Graphs

Store_Names

Store_Cities

City_Names

Product_Hierarchy

Product_Names

Workbook Statistics

100%

Module 1, Lesson 1: Data Cleaning and Preparation

Key Observations

- All datasets are **clean and consistent**; no missing or duplicate entries
- **Data relationships** are intact: every sale is linked to a **store, product, and city**
- Initial observations: some **stores and cities** already show higher total sales

Significance

- Ensures **accurate analysis** in later modules
- Provides early insights into **potential performance trends**

Module 1, Lesson 2: Data Analysis Using Pivot Tables

store name	Total Sales	Total Stock	Total Revenue	Average Sales
Currys (National Chain)	389	31473	1055.21	0.152
Darty	1901	45375	5080.29	0.644
DIGI	5538	133536	18146.50	0.675
Electro World (National Chain)	3675	153521	17880.98	0.382
Elettrodomestici Rossi	1247	71470	4545.47	0.246
Euronics Lisboa (National Chain)	398	30929	658.68	0.247
Grand Total	13148	466304	47367.13	0.438

City Name	Total Sales	Total Revenue	Average Sales
Edinburgh	5538	18146.50	0.675
Helsinki	3675	17880.98	0.382
London	1901	5080.29	0.644
Saint Petersburg	1247	4545.47	0.246
Vienna	787	1713.89	0.189
Grand Total	13148	47367.13	0.438

product_name	Total Sales	Total Revenue	Average Sales
Portable Heater	637	14378.98	3.283505155
Handheld Milk Frother	1218	2988.52	6.09
SolarGrillpulse	276	1123.66	1.15
PowerDryerflow	867	966.95	6.826771654
SolarBlenderlux	273	846.66	2.025
NeoHeatergen	101	806.8	0.512690355
SmartFridgewave	338	800.06	1.402489627
CyberTVwave	40	790.66	0.204081633
MaxDryerwave	150	736.6	0.75
TechVacuumflow	93	698.85	0.543859649
TechFridgematic	184	511.26	0.934010152
TechMixerflow	44	459.99	0.291390728
Hardwood Floor Cleaners	27	455.31	0.296703297
Smart Speakers (with voice assistants)	700	440.13	2.834008097
NanoFridgeflow	36	433.39	0.183673469
EcoVacuumcast	244	417.94	1.008264463
Immersion Blender	151	397.54	1.027210884
NanoGrillflow	1649	381.68	6.842323651
SolarOvenwave	117	352.14	0.606217617

Module 1, Lesson 2: Data Analysis Using Pivot Tables Results

1. Sales Data Analysis (by Store)

- **DIGI and Electro World** are the top-performing stores, with **5,538** and **3,675 sales** respectively
- Other stores show lower sales, highlighting **uneven performance across the chain**
- **Significance:** Identifies **where to focus inventory, marketing, and sales strategies**

Module 1, Lesson 2: Data Analysis Using Pivot Tables Results

2. Sales by City

- **Edinburgh and Helsinki** lead in sales and revenue, while cities like Vienna have lower sales
- **Significance:** Reveals **regional differences**, helping target **promotion and stock distribution**

Module 1, Lesson 2: Data Analysis Using Pivot Tables Results

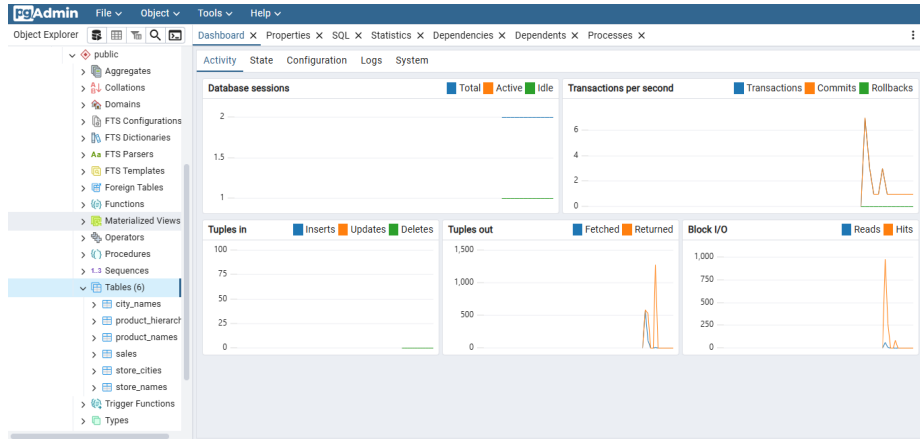
- **3. Product Performance Analysis**
- **NeoGrillflow, Handheld Milk Frother, and Portable Heater** are top-selling products
- **Significance:** Guides product stocking, pricing, and promotional decisions

Module 1, Lesson 2: Data Analysis Using Pivot Tables

Significance

- Highlights **top-performing stores and cities** that drive most of the company's revenue.
- Identifies **underperforming locations** where targeted marketing or training could improve performance.
- Reveals **key product categories** to prioritize for inventory planning and campaign focus.
- Establishes a **baseline understanding of overall performance** across stores, regions, and products.

Module 2, Lesson 1: Data Querying Using PostgreSQL



The screenshot shows the pgAdmin 4 Query Editor. The top toolbar includes buttons for query execution, refresh, and other utilities. The query editor contains the following SQL query:

```
1 SELECT * FROM sales LIMIT 5;
```

The bottom panel displays the Data Output tab, showing the results of the query in a table format:

	product_id	store_id	date	sales	revenue	stock	price	promo_type_1	promo_bin_1
	character varying	character varying	date	double precision	double precision	double precision	double precision	character varying	character varying
1	P0001	S0002	2017-02-01	0	0	8	6.25	PR14	[null]
2	P0001	S0012	2017-02-01	1	5.3	0	6.25	PR14	[null]
3	P0001	S0013	2017-02-01	2	10.59	0	6.25	PR14	[null]
4	P0001	S0023	2017-02-01	0	0	6	6.25	PR14	[null]
5	P0001	S0025	2017-02-01	0	0	1	6.25	PR14	[null]

Total rows: 5 of 5 Query complete 00:00:00.209 Ln 1, Col 29

The screenshot shows the pgAdmin 4 Query Editor. The top toolbar includes buttons for query execution, refresh, and other utilities. The query editor contains the following SQL query:

```
1 SELECT
2   p.product_id,
3   c.store_id,
4   c.city_id,
5   SUM(sl.sales) AS total_sales,
6   SUM(sl.revenue) AS total_revenue,
7   AVG(sl.price) AS average_price
8 FROM
9   sales sl
10 INNER JOIN product_hierarchy p ON sl.product_id = p.product_id
11 INNER JOIN store_cities c ON sl.store_id = c.store_id
12 GROUP BY
13   p.product_id,
14   c.store_id,
15   c.city_id;
```

The bottom panel displays the Data Output tab, showing the results of the query in a table format:

	product_id	store_id	city_id	total_sales	total_revenue	average_price
	character varying	character varying	character varying	double precision	double precision	double precision
1	P0001	S0002	C007	0	0	6.25
2	P0001	S0012	C005	1	5.3	6.25
3	P0001	S0013	C026	2	10.59	6.25
4	P0001	S0023	C008	0	0	6.25
5	P0001	S0025	C024	0	0	6.25
6	P0001	S0027	C022	0	0	6.25
7	P0001	S0040	C017	0	0	6.25
8	P0001	S0049	C031	0	0	6.25
9	P0001	S0050	C014	0	0	6.25
10	P0001	S0051	C027	0	0	6.25
11	P0001	S0055	C014	0	0	6.25
12	P0001	S0056	C015	1	5.3	6.25
13	P0001	S0062	C014	0	0	6.25
14	P0001	S0063	C033	0	0	6.25
15	P0001	S0066	C033	0	0	6.25

Total rows: 1000 of 13220 Query complete 00:00:00.279 Ln 16, Col 1

The screenshot shows the pgAdmin 4 Query Editor. The top toolbar includes buttons for query execution, refresh, and other utilities. The query editor contains the following SQL query:

```
1 SELECT
2   p.product_id,
3   c.store_id,
4   c.city_id,
5   SUM(sl.sales) AS total_sales,
6   SUM(sl.revenue) AS total_revenue,
7   AVG(sl.price) AS average_price
8 FROM
9   sales sl
10 INNER JOIN product_hierarchy p ON sl.product_id = p.product_id
11 INNER JOIN store_cities c ON sl.store_id = c.store_id
12 GROUP BY
13   p.product_id,
14   c.store_id,
15   c.city_id;
```

The bottom panel displays the Data Output tab, showing the results of the query in a table format:

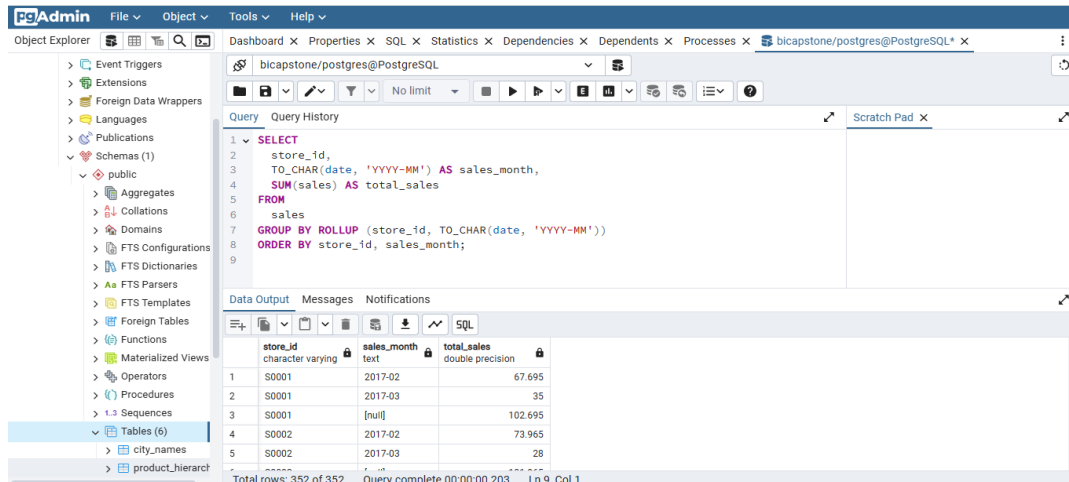
	product_id	store_id	city_id	total_sales	total_revenue	average_price
	character varying	character varying	character varying	double precision	double precision	double precision
1	P0001	S0002	C007	0	0	6.25
2	P0001	S0012	C005	1	5.3	6.25
3	P0001	S0013	C026	2	10.59	6.25
4	P0001	S0023	C008	0	0	6.25
5	P0001	S0025	C024	0	0	6.25
6	P0001	S0027	C022	0	0	6.25
7	P0001	S0040	C017	0	0	6.25
8	P0001	S0049	C031	0	0	6.25
9	P0001	S0050	C014	0	0	6.25
10	P0001	S0051	C027	0	0	6.25
11	P0001	S0055	C014	0	0	6.25
12	P0001	S0056	C015	1	5.3	6.25
13	P0001	S0062	C014	0	0	6.25
14	P0001	S0063	C033	0	0	6.25
15	P0001	S0066	C033	0	0	6.25

Total rows: 1000 of 13220 Query complete 00:00:00.279 Ln 16, Col 1

Module 2, Lesson 1: Data Querying Using PostgreSQL

- **Results**
- **Database Setup:** pgAdmin interface showing all tables in the BICapstone schema.
- **Data Verification:** SQL query checking one table (e.g., sales) confirms data is populated correctly.
- **Sales Performance Query:** SQL query joining tables to calculate total sales per store; results show **S0001 (DIGI)** and **S0002 (Electro World)** as top performers, while other stores contribute less.
- **Significance:**
- Setting up the PostgreSQL database provides a **structured environment** for querying and analyzing multi-table datasets.
- Ensures data integrity and consistency, which is **essential for all subsequent analyses** (ROLLUPs, cubes, and visualizations).

Module 2, Lesson 2: Data Analysis Using PostgreSQL



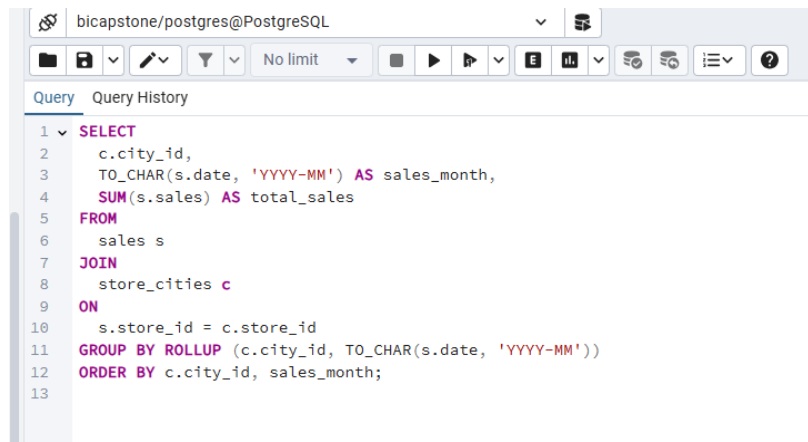
The screenshot shows the pgAdmin interface with the following components:

- Object Explorer:** A tree view on the left showing the database structure, including Schemas, Tables, and Views.
- Query Editor:** The central pane containing a SQL query. The query is a SELECT statement with a ROLLUP clause, grouping sales data by store and month.
- Data Output:** A table at the bottom displaying the results of the query. It has four columns: store_id, sales_month, and total_sales. The results show sales data for two stores (S0001 and S0002) across different months (2017-02 and 2017-03).

```
1 SELECT
2   store_id,
3   TO_CHAR(date, 'YYYY-MM') AS sales_month,
4   SUM(sales) AS total_sales
5 FROM
6   sales
7 GROUP BY ROLLUP (store_id, TO_CHAR(date, 'YYYY-MM'))
8 ORDER BY store_id, sales_month;
```

store_id	sales_month	total_sales
S0001	2017-02	67.695
S0001	2017-03	35
S0001	[null]	102.695
S0002	2017-02	73.965
S0002	2017-03	28

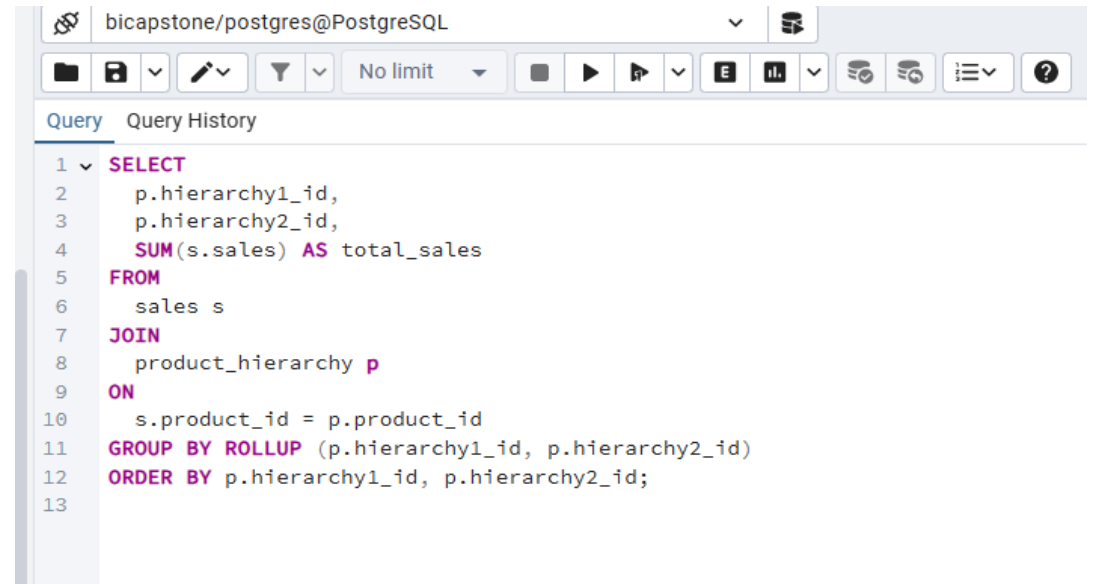
Total rows: 352 of 352 Query complete 00:00:00.203 Ln 9, Col 1



The screenshot shows the pgAdmin interface with the following components:

- Query Editor:** The central pane containing a SQL query. The query is a SELECT statement with a ROLLUP clause, grouping sales data by city and month.

```
1 SELECT
2   c.city_id,
3   TO_CHAR(s.date, 'YYYY-MM') AS sales_month,
4   SUM(s.sales) AS total_sales
5 FROM
6   sales s
7 JOIN
8   store_cities c
9 ON
10  s.store_id = c.store_id
11 GROUP BY ROLLUP (c.city_id, TO_CHAR(s.date, 'YYYY-MM'))
12 ORDER BY c.city_id, sales_month;
```



The screenshot shows the pgAdmin interface with the following components:

- Query Editor:** The central pane containing a SQL query. The query is a SELECT statement with a ROLLUP clause, grouping sales data by hierarchy and month.

```
1 SELECT
2   p.hierarchy1_id,
3   p.hierarchy2_id,
4   SUM(s.sales) AS total_sales
5 FROM
6   sales s
7 JOIN
8   product_hierarchy p
9 ON
10  s.product_id = p.product_id
11 GROUP BY ROLLUP (p.hierarchy1_id, p.hierarchy2_id)
12 ORDER BY p.hierarchy1_id, p.hierarchy2_id;
```

Module 2, Lesson 2: Data Analysis Using PostgreSQL

Data Output Messages Notifications			
SQL			
	hierarchy1_id character varying	hierarchy2_id character varying	total_sales double precision
1	H00	H0000	1380
2	H00	H0001	617
3	H00	H0002	849
4	H00	H0003	4268
5	H00	H0004	613.366
6	H00	[null]	7727.3660000000001
7	H01	H0105	175
8	H01	H0106	166
9	H01	H0107	374
10	H01	H0108	172
11	H01	[null]	887
12	H02	H0209	43.209999999999994
13	H02	H0210	182.30500000000004
14	H02	[null]	225.51500000000001
15	H03	H0311	17
Total rows: 23 of 23 Query complete 00:00:00.189 Ln 13, Col 1			

Data Output Messages Notifications			
SQL			
	city_id character varying	sales_month text	total_sales double precision
1	C002	2017-02	100
2	C002	2017-03	59
3	C002	[null]	159
4	C003	2017-02	32.84
5	C003	2017-03	14
6	C003	[null]	46.84
7	C004	2017-02	223.49
8	C004	2017-03	110.208
9	C004	[null]	333.698
10	C005	2017-02	54
11	C005	2017-03	35
12	C005	[null]	89
13	C006	2017-02	124
14	C006	2017-03	29
15	C006	[null]	153
Total rows: 100 of 100 Query complete 00:00:00.222 Ln 13, Col 1			

Module 2, Lesson 2: Data Analysis Using PostgreSQL

Significance:

- **ROLLUP and hierarchical summaries** reveal aggregate sales at multiple levels (city, store, product)
- Helps identify **high-performing regions and products**
- Supports **stock management and targeted marketing strategies**
- Provides insights to **optimize sales allocation and inventory planning**

Module 3, Lesson 1: Data Visualization Using Excel

1. Line Chart – Sales Performance Over Time

- Shows **daily sales fluctuations** from Jan 2 – Feb 16, 2017
- Peaks: Jan 21 (**613.95 sales**) and Feb 12 (**437.35 sales**)
- Dips highlight **days with low performance**
- **Significance:** Enables identification of **sales trends, seasonal spikes, and dips**, guiding inventory and promotion planning

2. Scatter Plot – Sales vs. Stock

- Most points concentrated in **low-to-mid sales with varying stock levels**
- Reveals **correlation between stock and sales**, highlighting overstocked or underperforming inventory
- **Significance:** Supports **stock optimization and demand planning**

Module 3, Lesson 1: Data Visualization Using Excel

3. Sunburst Chart – Sales Distribution by City

- Largest contributors: C007 (~1/3) and C031 (~1/3) of total sales
- Smaller contributors: C014, C022, C024
- **Significance:** Visualizes **regional sales contributions**, helping prioritize high-performing cities for marketing and resources

4. Bar Chart – Sales by Store

- DIGI and Electro World show higher sales; other stores are mostly low
- Highlights **store-level performance differences within chains**
- **Significance:** Guides **store-level performance evaluation and strategic planning**

Module 3, Lesson 2: Statistical Analysis

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	date	sales											
2	1/2/2017	0		SUMMARY OUTPUT									
3	1/2/2017	0											
4	1/2/2017	1		Regression Statistics									
5	1/2/2017	0		Multiple R	0.002547841								
6	1/2/2017	0		R Square	6.49149E-06								
7	1/2/2017	0		Adjusted R	-4.35157E-05								
8	1/2/2017	2		Standard E	2.805713082								
9	1/2/2017	0		Observatio	19999								
10	1/2/2017	0											
11	1/2/2017	0		ANOVA									
12	1/2/2017	0		df	SS	MS	F	Significance F					
13	1/2/2017	0		Regression	1	1.021877363	1.0218774	0.129811	0.71863165				
14	1/2/2017	0		Residual	19997	157416.9019	7.8720259						
15	1/2/2017	0		Total	19998	157417.9238							
16	1/2/2017	0											
17	1/2/2017	0		Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95%	Upper 95%		
18	1/2/2017	0		Intercept	-23.18840106	65.8376272	-0.3522059	0.724687	-152.23559	105.858784	-152.23559	105.85878	
19	1/2/2017	0		date	0.000554764	0.001539758	0.3602933	0.718632	-0.0024633	0.00357282	-0.0024633	0.0035728	
20	1/2/2017	0											
21	1/2/2017	0											
22	1/2/2017	0											

The regression coefficient for date is 0.00055, indicating that for each additional day, sales are expected to increase by a small fraction. The y intercept is -23.19 which means that sales are -23.19

The P-value for date is 0.718, which is greater than 0.05, which indicates that the relationship between date and sales is not statistically significant, so we cannot conclude that the sales change was impactful or had much of a significant

The R-squared value is 0.000006, meaning that almost none of the variation in sales is explained by date. Essentially, this shows us that date is not a good predictor of sales in this dataset.

Predicting for 2/5/2017	2/5/17 as a number
$y = ax + b$	42771
$y = 0.000554764 * 42771$	
23.72781104	
	write excel date as a number and plug into the regression formula for prediction of sales in the future

Module 3, Lesson 2: Statistical Analysis

Regression Analysis Overview:

- Objective: Forecast sales trends and assess relationships with date
- Method: Linear regression in Excel using **sales as dependent variable** and **date as independent variable**

Key Results:

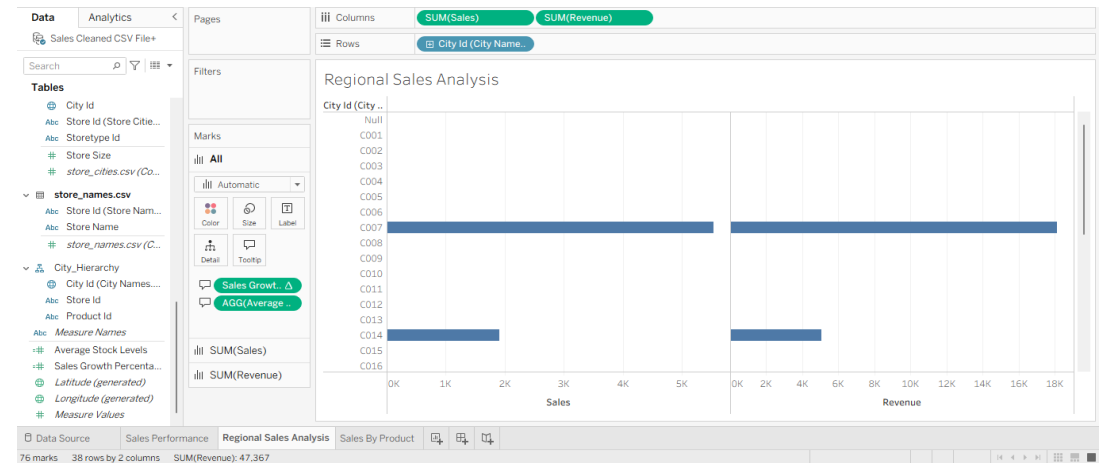
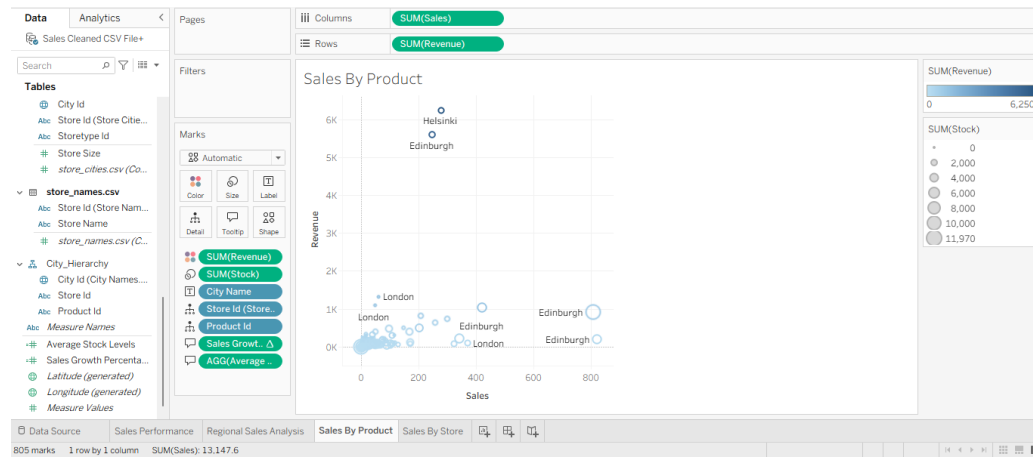
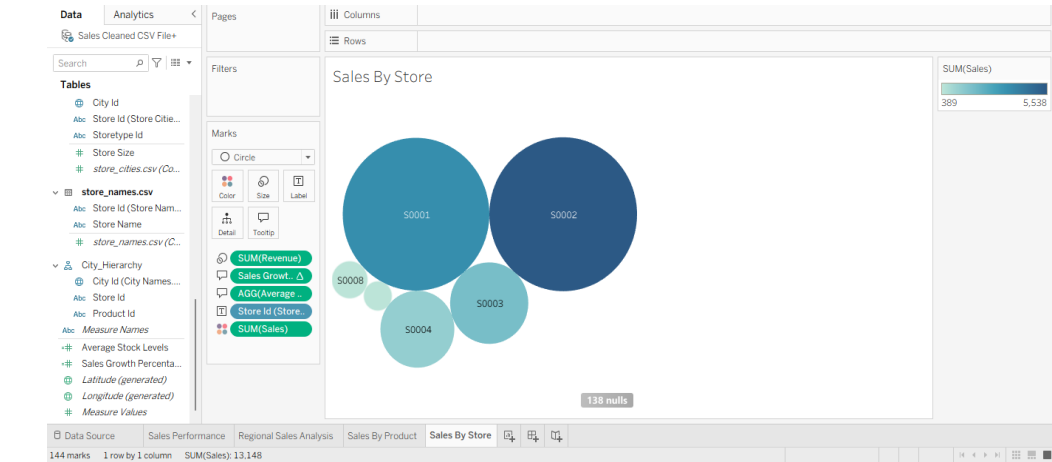
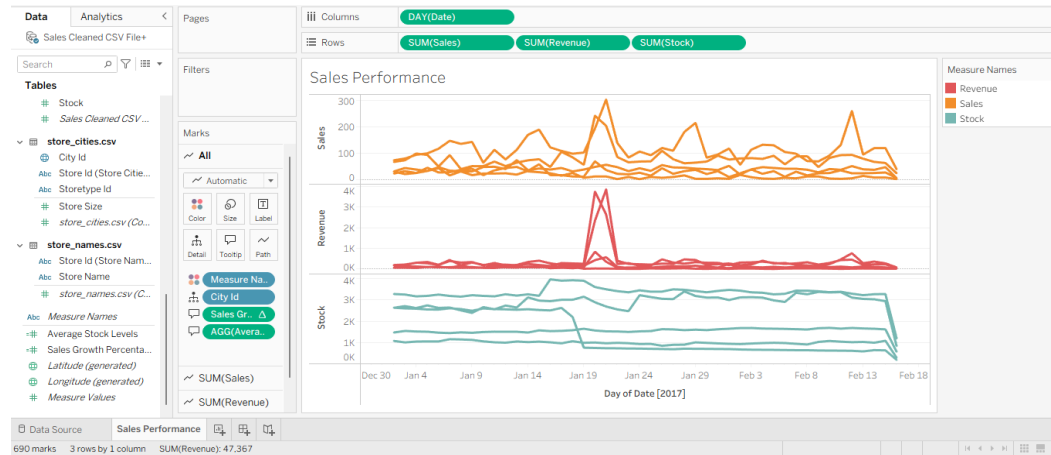
- Regression equation: $y = 0.00055 \times \text{date} - 23.19$
- $R^2 = 0.000006$ → Almost none of the variation in sales is explained by date
- P-value for date = 0.718 → Greater than 0.05, so date is **not a statistically significant predictor**

Module 3, Lesson 2: Statistical Analysis

Interpretation & Significance:

- **Sales are not strongly time-dependent** in this dataset; other factors likely drive sales variation
- Helps management understand that **daily sales fluctuations are not explained by linear trends over time**
- Supports decision-making by suggesting that **promotions, regional demand, or product factors** should be analyzed for forecasting instead

Module 4, Lesson 1: Basic Tableau Visualizations



Module 4, Lesson 1: Basic Tableau Visualizations

1. Line Chart – Sales Performance Over Time

- Shows **daily trends of sales, stock, and revenue**
- Highlights peaks (e.g., Jan 21 – 613 sales) and dips (e.g., Feb 16 – 82 sales)
- **Significance:** Quickly communicates **overall trends and fluctuations** for decision-making

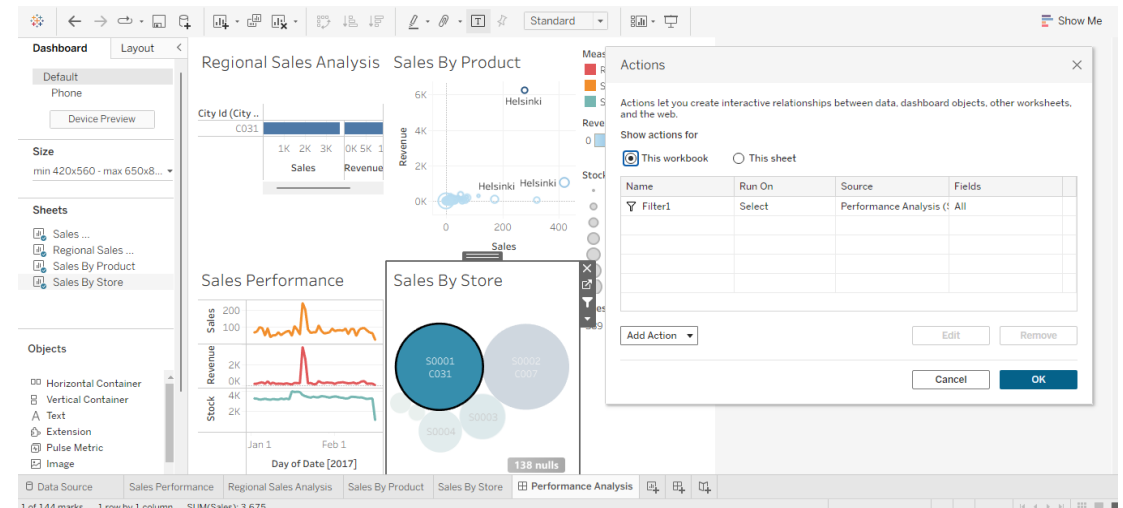
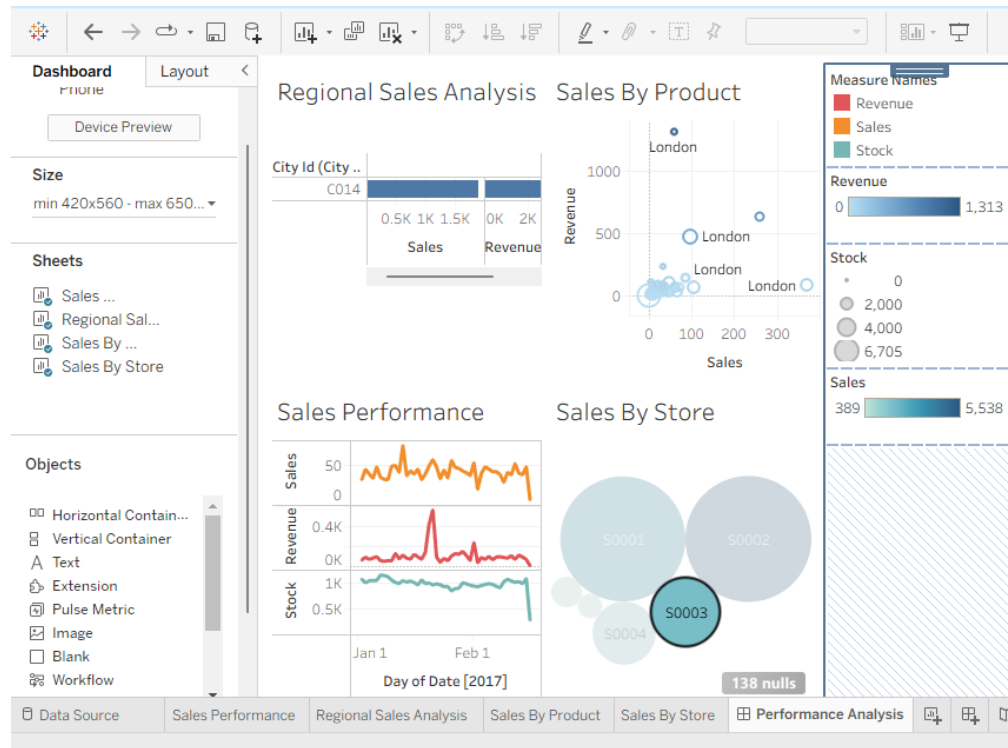
2. Bar Chart – Regional Sales Analysis

- Horizontal bars for **city-level sales and revenue**
- Top contributors: C007 (5,538 sales), C031 (3,675 sales); smaller contributors: C014, C022, C024
- **Significance:** Identifies **high-performing and underperforming regions** for targeted strategies

Module 4, Lesson 1: Basic Tableau Visualizations

- **3. Scatter Plot – Sales by Product**
- Bubbles show **sales vs. revenue** for products across cities
- Key observations:
 - Helsinki has highest revenue (6,250) from product P0129, store S0001
 - Edinburgh dominates sales (807 & 819 units for products P0282 & P0438)
- **Significance:** Reveals **which products and cities drive revenue**, aiding product and marketing decisions
- **4. Packed Bubble Chart – Sales by Store**
- Bubbles sized by **revenue and sales**
- Top stores: S001 (C031 – 17,881 revenue), S0002 (C007 – 18,147 revenue)
- Smaller stores highlighted as well
- **Significance:** Visualizes **store-level performance**, helping prioritize resources and promotions

Module 4, Lesson 2: Advanced Visualizations Using Tableau



Module 4, Lesson 2: Advanced Visualizations Using Tableau

Objective: Integrate previous analyses into an interactive dashboard for decision-making.

- **Key Features:**

- Combines multiple charts: **line, bar, scatter, and bubble charts**
- Interactive filters by **city, store, and product**
- Dynamic elements: **tooltips, calculated fields, and highlights**

Significance:

- Provides a **single interactive view** of sales performance
- Helps **quickly identify top-performing stores/products and underperforming regions**
- Facilitates **data-driven decisions** for inventory, marketing, and sales strategies



Discussion

Insights and Recommendations

- **Insights:**
- **Edinburgh** dominates product sales across multiple items → strong market presence.
- **Helsinki** leads in revenue for high-value products → strategic focus on premium items.
- **C024 and some smaller stores** underperform in sales and revenue → may need marketing support or inventory adjustments.
- **Product trends:** Certain products (e.g., P0129, P0282) consistently drive sales across multiple cities.
- **Sales fluctuations:** Daily trends show peaks and dips → useful for staffing, promotions, and inventory planning.
- **Recommendations:**
- **Targeted marketing** for underperforming regions (C024, smaller stores).
- **Stock optimization:** Prioritize high-demand products and cities with proven sales.
- **Promotions and campaigns** around peak sales periods identified in line charts.
- **Leverage dashboards** for ongoing monitoring and quick decision-making.



Conclusion

Summary

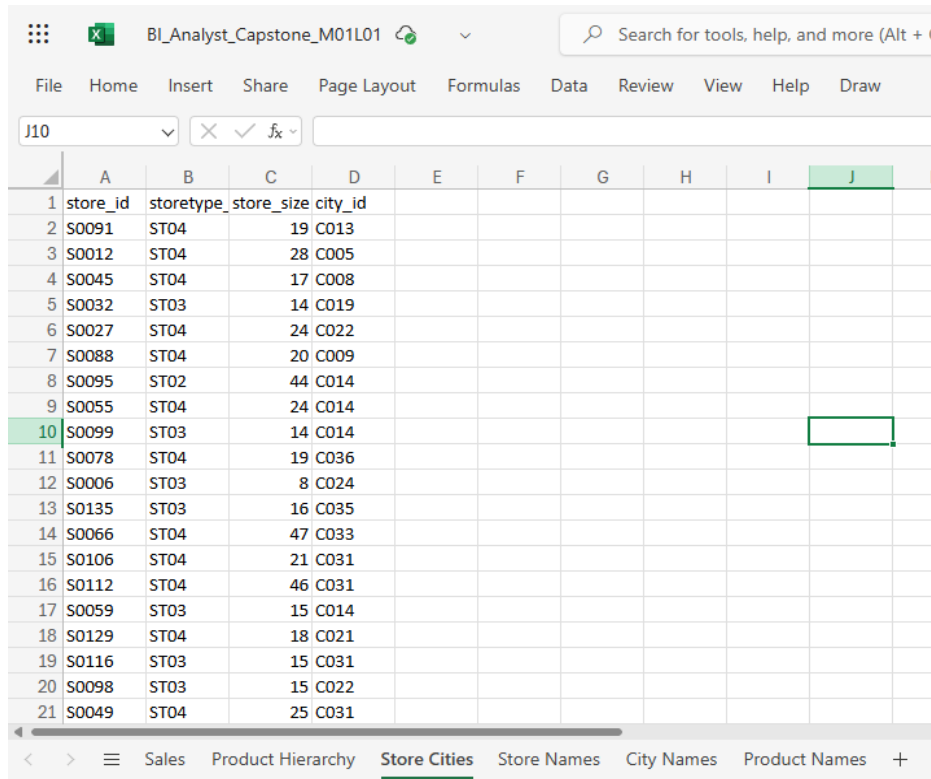
- **Comprehensive Analysis:** Cleaned, prepared, and analyzed sales, store, city, and product data.
- **Key Findings:**
 - Top-performing regions: **Edinburgh, Helsinki**
 - High-revenue products: **P0129, P0282**
 - Underperforming stores/regions: **C024 and smaller branches**
- **Visual Insights:** Pivot tables, Excel charts, and Tableau dashboards revealed **sales trends, product performance, and regional differences**.
- **Actionable Recommendations:** Optimize inventory, implement targeted marketing, and leverage dashboards for ongoing decisions.
- **Conclusion:** Data-driven insights enable the retail chain to **improve profitability, streamline operations, and enhance customer targeting**.



Appendix

Appendix #1

Include any relevant additional charts, worksheets, or tables that you may have created during the analysis phase



BI_Analyst_Capstone_M01L01

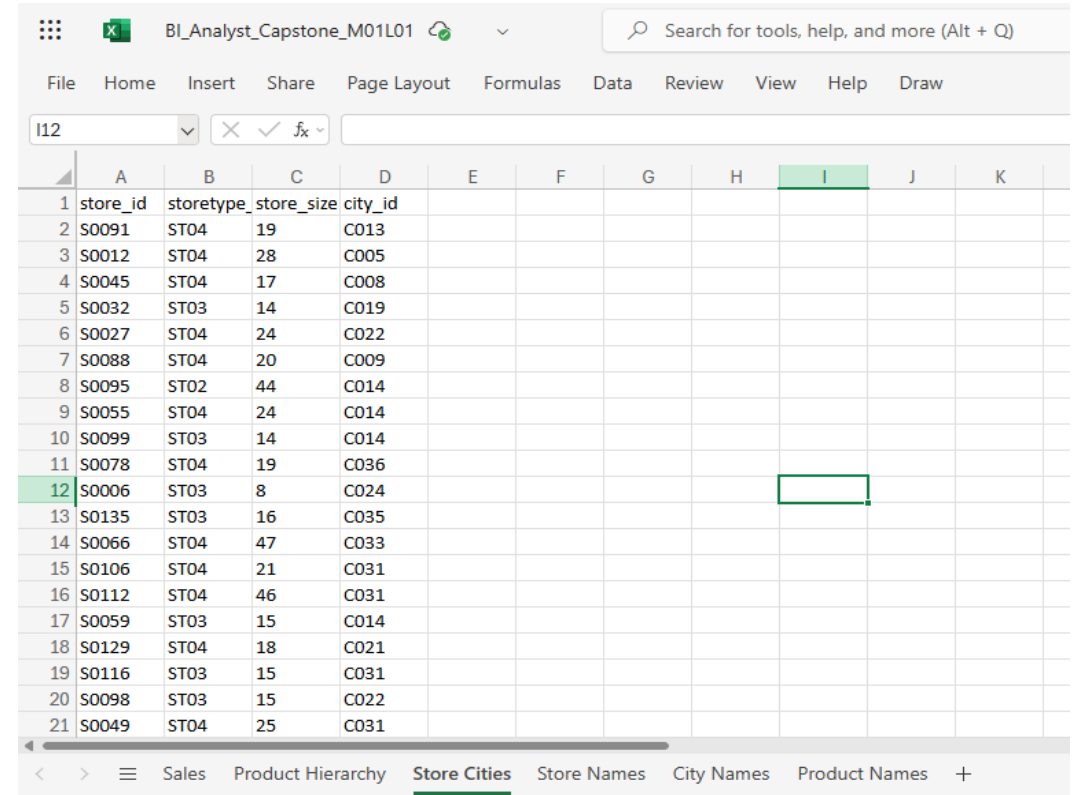
Search for tools, help, and more (Alt + Q)

File Home Insert Share Page Layout Formulas Data Review View Help Draw

J10

	A	B	C	D	E	F	G	H	I	J
1	store_id	storetype	store_size	city_id						
2	S0091	ST04	19	C013						
3	S0012	ST04	28	C005						
4	S0045	ST04	17	C008						
5	S0032	ST03	14	C019						
6	S0027	ST04	24	C022						
7	S0088	ST04	20	C009						
8	S0095	ST02	44	C014						
9	S0055	ST04	24	C014						
10	S0099	ST03	14	C014						
11	S0078	ST04	19	C036						
12	S0006	ST03	8	C024						
13	S0135	ST03	16	C035						
14	S0066	ST04	47	C033						
15	S0106	ST04	21	C031						
16	S0112	ST04	46	C031						
17	S0059	ST03	15	C014						
18	S0129	ST04	18	C021						
19	S0116	ST03	15	C031						
20	S0098	ST03	15	C022						
21	S0049	ST04	25	C031						

Sales Product Hierarchy Store Cities Store Names City Names Product Names +



BI_Analyst_Capstone_M01L01

Search for tools, help, and more (Alt + Q)

File Home Insert Share Page Layout Formulas Data Review View Help Draw

I12

	A	B	C	D	E	F	G	H	I	J	K
1	store_id	storetype	store_size	city_id							
2	S0091	ST04	19	C013							
3	S0012	ST04	28	C005							
4	S0045	ST04	17	C008							
5	S0032	ST03	14	C019							
6	S0027	ST04	24	C022							
7	S0088	ST04	20	C009							
8	S0095	ST02	44	C014							
9	S0055	ST04	24	C014							
10	S0099	ST03	14	C014							
11	S0078	ST04	19	C036							
12	S0006	ST03	8	C024							
13	S0135	ST03	16	C035							
14	S0066	ST04	47	C033							
15	S0106	ST04	21	C031							
16	S0112	ST04	46	C031							
17	S0059	ST03	15	C014							
18	S0129	ST04	18	C021							
19	S0116	ST03	15	C031							
20	S0098	ST03	15	C022							
21	S0049	ST04	25	C031							

Sales Product Hierarchy Store Cities Store Names City Names Product Names +

Workbook Statistics

Appendix #2

sales+



```
SET  
CREATE TABLE  
ALTER TABLE  
CREATE TABLE  
ALTER TABLE  
CREATE TABLE  
ALTER TABLE  
CREATE TABLE  
ALTER TABLE
```

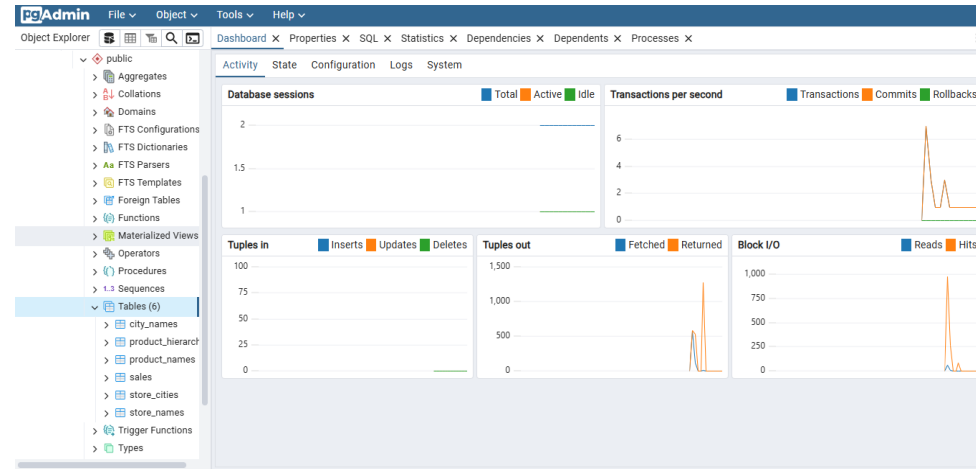
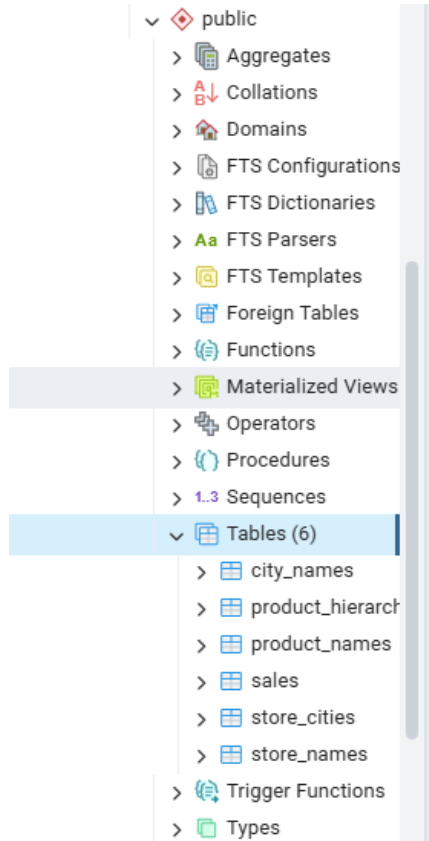
```
^  
CREATE DATABASE  
psql (14.17 (Ubuntu 14.17-0ubuntu0.22.04.1), serve  
r 13.2)  
You are now connected to database "bicapstone" as  
user "postgres".  
bicapstone=# \include '/home/project/BI-dump.sql'  
SET  
SET
```

Problems

theia@theiadocker-patrickchowt: /home/project X

```
COPY 37  
COPY 699  
COPY 699  
COPY 19999  
COPY 144  
COPY 144  
ALTER TABLE  
ALTER TABLE  
ALTER TABLE
```

Appendix #3



The SQL editor shows the following query:

```
SELECT * FROM sales LIMIT 5;
```

The results are displayed in a table with the following columns:

	product_id character varying	store_id character varying	date date	sales double precision	revenue double precision	stock double precision	price double precision	promo_type_1 character varying	promo_bin_1 character va
1	P0001	S0002	2017-02-01		0	0	8	6.25 PR14	[null]
2	P0001	S0012	2017-02-01		1	5.3	0	6.25 PR14	[null]
3	P0001	S0013	2017-02-01		2	10.59	0	6.25 PR14	[null]
4	P0001	S0023	2017-02-01		0	0	6	6.25 PR14	[null]
5	P0001	S0025	2017-02-01		0	0	1	6.25 PR14	[null]

Total rows: 5 of 5 Query complete 00:00:00.209 Ln 1, Col 29

Appendix #4

PgAdmin interface showing a query executed on the `product_hierarchy` table. The query is `SELECT * FROM product_hierarchy LIMIT 5;`. The results are displayed in a table with 10 columns: `product_id`, `product_length`, `product_depth`, `product_width`, `cluster_id`, `hierarchy1_id`, `hierarchy2_id`, `hierarchy3_id`, `hier`, and `cha`.

product_id	product_length	product_depth	product_width	cluster_id	hierarchy1_id	hierarchy2_id	hierarchy3_id	hier	cha
P0000	13.5	5	20	12	h00	H0004	H000401	H0C	
P0001	22	20	20	cluster_5	H01	H0105	H010501	H0I	
P0002	22	40	22	cluster_5	H03	H0315	H031508	H0G	
P0004	2	13	4	cluster_3	H03	H0314	H031405	H0C	
P0005	16	30	16	cluster_9	H03	H0312	H031211	H0C	

Total rows: 5 of 5 Query complete 00:00:00.189 Ln 1, Col 41

PgAdmin interface showing a query executed on the `store_cities` table. The query is `SELECT * FROM store_cities LIMIT 5;`. The results are displayed in a table with 4 columns: `store_id`, `storetype_id`, `store_size`, and `city_id`.

store_id	storetype_id	store_size	city_id
S0091	ST04	19	C013
S0012	ST04	28	C005
S0045	ST04	17	C008
S0032	ST03	14	C019
S0027	ST04	24	C022

Total rows: 5 of 5 Query complete 00:00:00.165 Ln 1, Col 36

PgAdmin interface showing a query executed on the `product_names` table. The query is `SELECT * FROM product_names LIMIT 5;`. The results are displayed in a table with 2 columns: `product_id` and `product_name`.

product_id	product_name
P0000	Bottom Freezer Refrigerator
P0001	Side-by-Side Refrigerator
P0002	Mini Fridge
P0004	Beverage Center
P0005	Wine Cooler

Total rows: 5 of 5 Query complete 00:00:00.203 Ln 1, Col 37

PgAdmin interface showing a query executed on the `store_names` table. The query is `SELECT * FROM store_names LIMIT 5;`. The results are displayed in a table with 2 columns: `store_id` and `store_name`.

store_id	store_name
S0005	MediaMarkt(National Chain)
S0036	Saturn
S0104	Euronics
S0068	FNAC
S0086	Darty

Total rows: 5 of 5 Query complete 00:00:00.201 Ln 1, Col 35